4. Algorithm

Binary Search Tree (BST) for Employee IDs

Step 1: Start the program.

Step 2: Initialize the BST by setting the root to NULL.

Step 3: Display the menu containing the following operations:

- Insert Employee ID
- Search Employee ID
- Delete Employee ID
- Display Employee IDs
- Exit

Step 4: Accept the user's choice.

Step 5: If the user selects Insert, create a new node and insert the employee ID according to the BST property.

Step 6: If the user selects Search, compare the given ID with the current node and move left or right until the ID is found or the search ends.

Step 7: If the user selects Delete, find the employee ID and delete its node. Handle the three cases: no child, one child, or two children.

Step 8: If the user selects Display, perform inorder traversal to display the IDs in ascending order.

Step 9: Repeat Steps 3–8 until the user selects Exit.

Step 10: Release the allocated memory and stop the program.